ROHS SOLDER PASTE

Complete Technical Guide

Alloys • Powder Types • Flux Classifications
Component Compatibility • Storage & Handling
Reflow Profiles • Quality Control

PCBA Manufacturing & Assembly Standards
IPC/J-STD-005 | IPC/J-STD-006 | RoHS Directive 2011/65/EU

1. OVERVIEW OF ROHS SOLDER PASTE

Solder paste is a homogeneous mixture of **lead-free solder alloy powder** and **flux medium**. It is the primary material for Surface Mount Technology (SMT) reflow soldering in RoHS-compliant PCBA manufacturing. The paste is stencil-printed onto PCB pads, components are placed by pick-and-place machines, and the assembly is heated in a reflow oven to form permanent metallurgical bonds.

Composition Breakdown:

- **Solder Powder (85–90% by weight):** SAC305, SAC387, SnCu, or specialty alloys
- **Flux Medium (10–15% by weight):** Activators, solvents, rheology modifiers, thickeners
- **Metal Load:** Typically 88–92% for standard applications; 85–87% for fine-pitch

Why RoHS Solder Paste?

- Lead content < 0.1% (1000 ppm) by weight — mandatory for EU market access
- Higher melting point (217–227°C) requires optimized thermal profiles
- Compatible with modern lead-free component terminations (Sn, NiPdAu, pure Sn)
- No-clean flux residues are generally benign and safe to leave on board

■ **Key Insight:** The quality of a solder paste is determined by three factors: **alloy composition** (melting, wetting, reliability), **powder particle size** (printability, response-to-pause), and **flux chemistry** (activity, residue, voiding performance).

2. ROHS SOLDER PASTE ALLOY TYPES

2.1 SAC305 (Sn96.5 / Ag3.0 / Cu0.5)

SAC305 is the global industry standard for RoHS solder paste. The silver content (3%) provides mechanical strength and thermal fatigue resistance, while copper (0.5%) improves wetting and reduces copper pad dissolution during reflow.

Technical Specifications:

- **Melting Point:** 217–220°C (eutectic range)
- **Recommended Peak Reflow:** 245–250°C
- **Time Above Liquidus (TAL):** 30–90 seconds
- **Joint Appearance:** Matte, grainy finish (normal for lead-free)
- **Voiding Tendency:** Moderate — requires optimized flux and profile
- **Copper Dissolution:** 2.1× higher than SnPb — monitor thin copper pads
- **Drop Shock:** Inferior to SnPb — consider SAC105 for portable devices

Best Applications:

- High-reliability consumer electronics (TVs, computers, appliances)
- Automotive PCBA (Tesla, Zoox compliance requirements)
- BGA, CSP, QFN area array packages
- General SMT assembly on standard FR-4 PCBs

■■ **Process Caution:** SAC305 exhibits **hot tear sensitivity** in large thermal mass joints and higher shrinkage during solidification. Preheat zones must be carefully tuned to prevent component tombstoning and solder balling.

2.2 SAC387 (Sn95.5 / Ag3.8 / Cu0.7)

SAC387 was one of the earliest RoHS alloys adopted in Japan and high-reliability sectors. The higher silver content (3.8%) improves mechanical strength and creep resistance but increases cost and can reduce drop-shock performance compared to SAC305.

- **Melting Point:** 217°C (eutectic)
- **Peak Reflow:** 245–250°C
- **Cost:** ~15% higher than SAC305 due to extra silver
- **Use Case:** High-stress environments, early RoHS adopters, Japanese OEMs

2.3 SAC105 (Sn98.5 / Ag1.0 / Cu0.5)

SAC105 is a **low-silver** alloy designed specifically for **drop-shock resistance**. Portable electronics (smartphones, tablets) benefit from SAC105's superior mechanical impact performance, though thermal cycling endurance is reduced compared to SAC305.

- **Melting Point:** 217°C
- **Peak Reflow:** 240–245°C (slightly lower than SAC305)
- **Drop Shock:** 2–3× better than SAC305
- **Thermal Cycling:** Inferior to SAC305 — avoid in automotive
- **Use Case:** Mobile devices, handheld consumer electronics

2.4 SnCu & SnCu+Ni (K100) — Paste Form

While **SnCu** is primarily a wave/bar solder alloy, paste formulations exist for cost-sensitive applications. The higher melting point (227°C) and slower wetting make it challenging for reflow, but it offers the lowest material cost.

- **Melting Point:** 227°C
- **Peak Reflow:** 250–260°C
- **Wetting:** Slower than SAC — requires strong flux activity
- **Cost:** ~50% of SAC305 price
- **Use Case:** Cost-driven consumer goods, non-critical assemblies

2.5 Specialty & Low-Temperature Alloys

SnBi58 Paste (Low Temperature)

With a melting point of only **138°C**, SnBi58 is used for temperature-sensitive components (polymer-based devices, LEDs, flex circuits). ■■ **CRITICAL:** Never mix with leaded processes — SnBi + Pb forms a 96°C eutectic that causes catastrophic joint failure.

- **Peak Reflow:** 165–175°C
- **Use Case:** LED assemblies, flex circuits, plastic components

SnBiAg (Near-SAC Alternative)

SnBiAg (~91% Sn, 4.8% Bi, 3.4% Ag) offers a compromise with melting point ~213°C. It reduces peak temperature requirements while maintaining reasonable mechanical properties. Still requires lead-free process discipline.

Innolot / HT Alloys

High-reliability alloys like **Innolot** (SnAgCu + Sb, Ni, Bi dopants) are designed for extreme automotive under-hood applications. They offer superior thermal cycling and creep resistance but at 4–5× the cost of SAC305.

3. SOLDER POWDER PARTICLE SIZE (TYPES)

Solder powder is classified by particle size distribution per **IPC/J-STD-005**. The type number determines stencil aperture design, printability, and minimum component pitch capability.

Type	Particle Size (µm)	Min Stencil Aperture	Best For
Type II	45–75	> 250µm	Coarse SMT, large pads, through-hole paste
Type III	25–45	200–250µm	Standard SMT, 0402/0603, SOIC, QFP
Type IV	20–38	150–200µm	Fine pitch QFP, 0201, small QFN
Type V	15–25	100–150µm	Ultra-fine 01005, micro-BGA, CSP
Type VI	5–15	< 100µm	Jet printing, 008004, advanced SiP
Type VII	2–11	< 80µm	Next-gen jet dispensing, wafer bumping

Table 1: IPC/J-STD-005 Solder Powder Classification

Selection Guidelines:

- **Type III** is the workhorse for 95% of SMT applications
- **Type IV** required for 0201 passives and fine-pitch QFP (< 0.5mm pitch)
- **Type V/VI** used with jet dispensers (no stencil) for ultra-miniature components
- **Smaller particles = higher surface oxidation** → shorter stencil life, more voiding

■■ **Storage Note:** Type V, VI, and VII powders have exponentially higher surface area and oxidize faster. These pastes require stricter refrigeration (0–5°C) and shorter shelf life (3 months vs 6 months for Type III).

4. FLUX CLASSIFICATIONS & CHEMISTRY

Flux is the chemical heart of solder paste. It cleans oxides, promotes wetting, and protects liquid solder from re-oxidation during reflow. RoHS fluxes must be compatible with lead-free temperatures (up to 260°C) without breaking down or leaving corrosive residues.

4.1 IPC Flux Activity Levels

Code	Activity	Halide Content	Residue	Use Case
L0	Low	0.0%	Non-corrosive, benign	No-clean, high-impedance circuits
L1	Low	< 0.5%	Non-corrosive, benign	No-clean, general SMT
M0	Moderate	0.0%	May be corrosive if not cleaned	Water-washable, high reliability
M1	Moderate	0.5–2.0%	Corrosive if not cleaned	RMA, water-wash, military
H0	High	0.0%	Corrosive — must clean	Difficult-to-solder surfaces
H1	High	> 2.0%	Corrosive — must clean	RA flux, plumbing (NOT PCBA)
ORH0	Organic High	0.0%	Non-corrosive, benign	No-clean lead-free, SAC305
ORH1	Organic High	< 0.5%	Non-corrosive, benign	No-clean lead-free, SAC305

Table 2: IPC/J-STD-004 Flux Classification for Electronics

4.2 Flux Chemistry Types

No-Clean (ORH0 / ORH1)

The dominant choice for modern SMT. Residues are benign, non-corrosive, and electrically safe. Saves washing costs and water consumption. **Limitation:** Weak activity — may struggle with heavily oxidized components or OSP (Organic Solderability Preservative) PCBs.

Water-Soluble (WS / M0-M1)

Strong activators provide excellent wetting on difficult surfaces. Residues are hygroscopic and corrosive — **must be washed** with DI water after reflow. Common in military, medical, and high-reliability applications where zero residue is required.

Rosin-Based (RMA / RA)

Traditional flux using natural rosin (colophony). RMA (Rosin Mildly Activated) is safe for most electronics. RA (Rosin Activated) is too aggressive for PCBA and reserved for plumbing or repair of heavily oxidized parts.

5. COMPONENT TERMINATION COMPATIBILITY

RoHS solder paste must wet and bond to component terminations. The plating material on component leads and pads determines compatibility, joint reliability, and potential failure modes.

Termination	Compatibility	Wetting	Concerns
Pure Tin (Sn)	■ Excellent	Very Good	Tin whisker growth (long-term reliability)
Matte Sn	■ Excellent	Excellent	Lower whisker risk than bright Sn
NiPdAu	■ Excellent	Excellent	Gold embrittlement if > 3μm Au
SnBi	■■ Caution	Good	Low melting — reflow can remelt termination
SnAg	■ Excellent	Excellent	Standard RoHS termination
SnCu	■ Good	Good	Slightly slower wetting than SAC
OSP (Cu)	■ Good	Moderate	Requires active flux; short shelf life
Immersion Ag	■ Excellent	Excellent	Silver migration risk in high humidity
Immersion Sn	■ Good	Good	Whisker risk; IMC growth over time
ENIG (Ni/Au)	■ Excellent	Excellent	Black pad syndrome (brittle Ni-P layer)
HASL SnPb	■ Incompatible	N/A	Mixing leaded and lead-free is prohibited

Table 3: Component Termination Compatibility with RoHS Solder Paste

Key Compatibility Rules:

- **Pure Sn terminations:** Most common, good wetting, but monitor for tin whiskers in high-stress applications
- **NiPdAu:** Premium termination — excellent solderability, no whiskers, but expensive
- **OSP:** Cost-effective but requires paste with strong activators and careful storage
- **ENIG:** Widely used but risk of 'black pad' — brittle fracture at Ni/solder interface
- **NEVER mix SnPb HASL with RoHS paste** — creates non-compliant product and unreliable joints

6. STORAGE, HANDLING & SHELF LIFE

6.1 Refrigeration Requirements

Solder paste is a chemically active system. Elevated temperatures accelerate flux degradation, oxidation, and viscosity changes. Proper cold storage is critical.

- **Storage Temperature:** 0–10°C (32–50°F) — standard refrigerator, NOT freezer
- **Freezing Risk:** Below 0°C can separate flux from powder, ruining rheology
- **Shelf Life (Sealed):** 6 months at 0–10°C for Type III; 3 months for Type V/VI
- **Shelf Life (Opened):** 4–8 hours at room temperature after opening jar
- **Transport:** Use insulated cold packs; avoid direct sunlight exposure

6.2 Thawing Procedure

Improper thawing causes condensation (moisture ingress) and temperature shock. Follow this sequence:

1. Remove from refrigerator and place at **room temperature (20–25°C)**
2. Allow **minimum 4 hours** to reach ambient temperature naturally
3. **DO NOT force-thaw** with heat guns, ovens, or hot water baths
4. Wipe any external condensation from jar before opening
5. Record thaw time on jar label (traceability requirement)

6.3 Stencil Printing Life

Once on the stencil, solder paste begins to dry out due to solvent evaporation and flux activation.

- **Stencil Life:** 4–8 hours at 20–25°C, 40–60% RH
- **Response-to-Pause:** Paste must recover viscosity after 1-hour idle; test before resuming
- **Scrap paste** that exceeds stencil life — do NOT return to original jar
- **Mixing old + new paste** is prohibited — contaminates fresh batch

■■ **FIFO Rule:** First-In-First-Out inventory management is mandatory. Use oldest paste first. Expired paste must be discarded or returned to supplier for metal reclamation. Never use paste past its expiration date — voiding and non-wetting defects will increase dramatically.

7. REFLOW PROFILE FOR ROHS SOLDER PASTE

Lead-free reflow requires higher peak temperatures and tighter process control than SnPb. The profile must balance complete wetting against component and PCB thermal limits.

7.1 Standard SAC305 Reflow Profile

Zone	Temperature	Time / Rate	Purpose
Preheat	25 → 150°C	60–120 sec 1–3°C/sec	Evolve solvents, activate flux gently
Soak / Thermal Equalization	150 → 180°C	60–120 sec	Equalize temperature across board; complete flux activation
Ramp to Peak	180 → 245°C	1.5–2.5°C/sec	Rapid heating to liquidus; minimize thermal shock
Reflow (Above Liquidus)	> 217°C	30–90 sec	Solder melts, wets pads and leads, forms joints
Peak Temperature	245–250°C	10–20 sec at peak	Maximum wetting; IMC formation
Cooling	250 → 100°C	< 4°C/sec	Solidify joints; prevent grain growth and pad lifting

Table 4: Recommended SAC305 Reflow Profile Parameters

7.2 Critical Profile Rules

- **Max ramp rate:** 3°C/sec to prevent component cracking and delamination
- **Min TAL:** 30 seconds above 217°C to ensure complete wetting of large thermal masses
- **Max TAL:** 90 seconds to prevent excessive IMC growth and PCB degradation
- **Peak limit:** 260°C absolute maximum for standard components; 245°C for moisture-sensitive
- **Cooling rate:** 2–4°C/sec optimal; too slow causes coarse grain structure

7.3 Nitrogen Atmosphere Benefits

Using **N2 atmosphere** (100–1000 ppm O2) in the reflow oven significantly improves lead-free soldering:

- Reduces oxide formation on solder powder and component terminations
- Improves wetting by 20–40% — critical for OSP and difficult surfaces
- Reduces voiding in BGA and QFN thermal pads
- Allows use of weaker (lower activity) no-clean fluxes
- **Cost:** Adds ~5–10% to process cost but reduces defect rates significantly

8. QUALITY CONTROL & COMMON DEFECTS

8.1 Voiding Analysis

Voiding (air pockets trapped in solder joints) is more prevalent in lead-free soldering due to higher surface tension and different solidification behavior. X-ray inspection is the primary method for measuring voids in hidden joints (BGA, QFN).

- **Acceptable Void Area:** < 25% of joint area per IPC-A-610 Class 2
- **Acceptable Void Area:** < 15% of joint area per IPC-A-610 Class 3 (high reliability)
- **Common Causes:** Insufficient flux, excessive peak temperature, poor pad design
- **Mitigation:** Optimize flux chemistry, use N2 atmosphere, design proper thermal pad apertures

8.2 Common RoHS Paste Defects

Tombstoning (Drawbridging)

Passive components (resistors, capacitors) stand on end like a tombstone. Caused by unequal wetting forces on the two terminations during reflow. SAC305's higher surface tension exacerbates this vs SnPb.

→ **Fix:** Optimize pad design (equal size), reduce stencil aperture offset, slow cooling rate

Solder Balling

Small solder spheres scattered around the joint. Caused by paste spatter from rapid solvent evaporation or oxidation of paste particles.

→ **Fix:** Extend preheat soak time, reduce ramp rate, verify paste storage conditions

Non-Wetting / Dewetting

Solder fails to spread across the pad surface, leaving exposed copper. Caused by oxidation, insufficient flux activity, or contaminated surfaces.

→ **Fix:** Check PCB shelf life, verify flux activity level, increase preheat, use N2

Head-in-Pillow (HiP)

BGA solder ball partially collapses but does not fully bond to PCB pad — looks like a head resting on a pillow. Caused by PCB or BGA warpage during reflow, or oxidation preventing metallurgical bonding.

→ **Fix:** Reduce peak temperature, extend TAL, improve board support, verify BGA storage

Hot Tears / Shrinkage Cracks

Visible cracks in large solder joints (connectors, transformers) after solidification. Caused by SAC305's 4% volume shrinkage during phase change and thermal contraction.

→ **Fix:** Switch to SnCu+Ni for wave soldering, optimize cooling rate, redesign joint geometry

■ **Best Practice:** Implement Statistical Process Control (SPC) on reflow oven temperatures. Profile boards with thermocouples daily. Paste performance degrades predictably — track defect PPM trends to catch paste aging before catastrophic failure.

9. SOLDER PASTE SELECTION DECISION MATRIX

Use this matrix to select the optimal solder paste for your specific application requirements.

Application	Recommended Alloy	Powder Type	Flux	Notes
General Consumer SMT	SAC305	Type III	ORH1 No-Clean	Workhorse choice
Automotive (Under-hood)	Innolot / SAC305+	Type III/IV	ORH1 No-Clean	High thermal cycling required
Smartphone / Tablet	SAC105	Type IV/V	ORH0 No-Clean	Drop-shock priority
LED Lighting	SnBi58	Type III	ORH1 No-Clean	Low temp protects LEDs
Flex / Rigid-Flex	SnBiAg	Type IV	ORH1 No-Clean	Low temp prevents flex damage
High-Volume Low-Cost	SnCu	Type III	M1 Water-Wash	Cheapest option; needs wash
Medical / Military	SAC305 / SAC387	Type III	M1 Water-Wash	Zero residue mandatory
BGA / CSP Arrays	SAC305	Type IV	ORH1 No-Clean	N2 recommended; voiding control
0201 / 01005 Passives	SAC305	Type V/VI	ORH0 No-Clean	Jet printing or ultra-fine stencil
Power Electronics	SAC305 / SnAgCu+	Type III	ORH1 No-Clean	High Ag for creep resistance

Table 5: Application-Based Solder Paste Selection Matrix

10. QUICK REFERENCE SUMMARY

Alloy Selection:

→ **SAC305** — 95% of applications; best balance of cost, performance, availability

→ **SAC105** — Drop-shock critical (mobile devices)

→ **SnBi58** — Temperature-sensitive only (LEDs, flex)

→ **SnCu** — Lowest cost; acceptable for non-critical consumer goods

Powder Type:

→ **Type III** — Standard SMT (0402, 0603, SOIC, QFP)

→ **Type IV** — Fine pitch (0201, 0.5mm QFP, small QFN)

→ **Type V/VI** — Ultra-fine (01005, jet dispensing, SiP)

Flux Selection:

→ **ORH1 No-Clean** — Standard for SAC305 reflow

→ **M1 Water-Wash** — High reliability; mandatory cleaning step

→ **ORH0** — Ultra-low residue for sensitive impedance circuits

Storage Rules:

→ **0–10°C** refrigerated, sealed, FIFO inventory

→ **4-hour minimum thaw** at room temperature before use

→ **4–8 hour stencil life** at room temperature

→ **Never return** used paste to original container

Reflow Essentials:

→ **Peak:** 245–250°C for SAC305

→ **TAL:** 30–90 seconds above 217°C

→ **Ramp:** 1.5–3°C/sec max

→ **N2 atmosphere** recommended for BGA and difficult surfaces

References & Standards:

- IPC/J-STD-005: Requirements for Soldering Pastes
- IPC/J-STD-006: Requirements for Electronic Grade Solder Alloys
- IPC/J-STD-004: Requirements for Soldering Fluxes
- IPC-A-610: Acceptability of Electronic Assemblies
- RoHS Directive 2011/65/EU (Restriction of Hazardous Substances)
- IPC-1752A: Materials Declaration Management
- Indium Corporation, Alpha Assembly, Kester Technical Datasheets

End of Document — RoHS Solder Paste Complete Technical Guide